



Fig. 19: The 12.6cm (5-inch) gun firing in a destroyer



Fig. 20: Harpoon missile firing from an Aegis cruiser

navigation system to fly the trajectory entered into its guidance system by its FCS. Upon nearing the target, say, 40km, it uses its radar seeker and finds its intended target.

Tackling asymmetric threats.

The general perception of US Naval planners was that elements of a CSG are slightly vulnerable to anti-ship missiles. But sub-surface and surface threats were considered as well covered due to CSG's superior radar/sonar and offensive capabilities. They expected that such long-, medium- and close-range threats would be neutralised by these capabilities before these would indeed become an actual threat.

But the art of war is not static; it is highly dynamic. New types of threats emerge at any point of time. On October 12, 2000, an American Aegis destroyer called USS Cole felt the brunt of such a new threat. This new threat was suicide boats. USS Cole docked in Aden harbour for refuelling. After some time, a small boat approached the destroyer

at high speed. Before anyone knew why it was approaching at such high speed, the boat rammed the ship and exploded. A huge hole was created on the hull and 17 US Navy personnel were killed, almost sinking the ship. Later investigations and analyses showed that the blast was caused by Al Qaeda suicide terrorists.

After the attack on USS Cole, gravity of the emerging scenario was realised. These boats presented themselves as small-size fast-moving targets. Such targets are difficult to shoot down by the 12.6cm (5-inch) guns of Aegis ships. Moreover, technologies are already in

place for remotely piloting such boats. So dozens of such boats can simultaneously engage each escorting Aegis ship and damage it. Such attacks are called swarming attacks (like a group of honey bees attacking an animal). It was realised that if such a swarming boat attack is successful, Aegis escorts will probably sink progressively, thus leaving the carrier vulnerable and on its own.

Such boats could only be shot down by manually-fired light cannons (type of heavy machine guns). Normal machine guns fire slugs, whereas, these light cannons fire explosive shells. But with both the ships and fast-moving target boats being heaved by waves, these machine guns were not efficient.

When the ship heaves up/down, the shooter has to tilt the gun downwards/upwards to hit the target. It is difficult to shoot a single target, but it is practically impossible to shoot down a swarm of suicide boats. So a sophisticated weapon system was needed, and that resulted in the

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Plot No. EL-60, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2787 3300 (Board), 2787 3311-16 (Sales)
Fax : 0091-22-2787 3310, 2787 3330
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